

Vol 10 Chapter 7 — Differential Geometry

Keeper (author pass — deep math/physics revision)

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Chapter 7 — Differential Geometry

Level 1 — one sentence

Differential geometry treats smooth manifolds, tangent bundles, connections, and curvature — underlying general relativity (Riemannian geometry) and gauge theory (bundle connections) — and the BST substrate D_{IV}^5 is a Kähler-Einstein manifold whose Bergman metric has curvature that gives Newton's gravitational constant G at 0.07% precision (Vol 4 Ch 1).

Level 2 — graduate-physicist precision

7.1 Smooth manifolds

An n -dim smooth manifold M : topological space locally homeomorphic to $\mathbb{R}^n$ with smooth transition functions. Examples: $\mathbb{R}^n$, S^n , T^n , Lie groups, projective spaces.

Tangent space $T_p M$ at p : vector space of tangent vectors. Tangent bundle $TM = \bigsqcup_p T_p M$.

Smooth functions $C^\infty(M)$; vector fields = sections of TM .

7.2 Riemannian manifolds

A Riemannian metric g_{ij} : smooth choice of inner product on each $T_p M$. Gives infinitesimal lengths $ds^2 = g_{ij} dx^i dx^j$, angles, volumes.

Levi-Civita connection ∇ : unique torsion-free connection compatible with metric. Parallel transport along curves.

Riemann curvature tensor R^i_{jkl} : how vectors fail to return to themselves after parallel transport around a loop. Quantifies intrinsic curvature.

Ricci tensor $R_{ij} = R^k_{ikj}$; scalar curvature $R = g^{ij} R_{ij}$.

7.3 Kähler manifolds

A Kähler manifold is a complex manifold with a Hermitian metric whose Kähler form ω is closed: $d\omega = 0$.

Equivalent: there exists a Kähler potential $K(z, \bar{z})$ with $g_{i\bar{j}} = \partial_i \partial_{\bar{j}} K$.

Properties: rich complex-analytic structure; many cohomology groups; Hodge decomposition.

7.4 D_{IV}^5 as Kähler-Einstein manifold

The bounded symmetric domain D_{IV}^5 is Kähler with Bergman metric — the metric induced by the Bergman kernel:

$$g_{i\bar{j}} = \partial_i \partial_{\bar{j}} \log K(z, \bar{z})$$

The Bergman metric on D_{IV}^5 is Kähler-Einstein (Ricci tensor proportional to metric). It's the substrate's natural geometric structure.

7.5 Bergman curvature and Newton's G

Volume 4 Chapter 1 (BST framework): Newton's gravitational constant G is derivable from the Bergman curvature of D_{IV}^5 :

$$G_{\text{BST}} \approx 6.67 \times 10^{-11} \text{ N m}^2/\text{kg}^2$$

with 0.07% precision. The substrate's natural curvature scale at the Bergman metric sets the gravitational coupling.

This is one of BST's headline derivations. No fit; G derived from substrate geometry.

7.6 K-audit anchors

- Volume 4 Chapter 1: Newton's G from Bergman curvature
- Volume 11 Chapter 2: Bergman kernel + metric
- SP-19b AB-10 (task #133): Newton's G derivation

Level 3 — 5th-grader accessibility

Differential geometry: math of smooth curved spaces. Riemannian manifolds have a metric (way to measure lengths/angles); curvature measures how “non-flat” they are. Kähler manifolds are complex-Riemannian — richer structure. BST's D_{IV}^5 is a Kähler-Einstein manifold with the Bergman metric built from the Bergman kernel. The Bergman curvature is what gives Newton's $G \approx 6.67 \times 10^{-11} \text{ N m}^2/\text{kg}^2$ at 0.07% precision — derived, not fit. One of BST's headline results.

What comes next

Chapter 8 develops representation theory.

Where to look this up

- Spivak, *A Comprehensive Introduction to Differential Geometry*
- Mok, *Metric Rigidity Theorems on Hermitian Locally Symmetric Manifolds*
- BST Vol 4 Ch 1; Vol 11 Ch 2; SP-19b AB-10